

# Project Initialization and Planning Phase

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|---------------|-----------------------------------------------------------------|
| Date          | 20 Sep 2026                                                     |
| Team ID       | Master of Computer Applications                                 |
| Project Title | <b>Online Learning Engagement and Student Performance Analy</b> |
| Maximum Marks | 3 Marks                                                         |

## Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

| Project Overview  |                                                                                                                                                                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | Develop an analytical and predictive framework to monitor student engagement metrics in online learning environments and evaluate their direct impact on academic performance.                                                                                                   |
| Scope             | Focuses on processing student interaction logs, tracking online participation rates, evaluating assessment scores, building predictive models for student performance, and visualizing insights via an interactive dashboard. Excludes offline traditional classroom activities. |
| Problem Statement |                                                                                                                                                                                                                                                                                  |
| Description       | In online learning environments, educators struggle to track student engagement in real time, leading to delayed interventions for disengaged or struggling students.                                                                                                            |
| Impact            | Unidentified disengagement leads to higher course dropout rates, lower overall academic achievement, and inefficient resource allocation for academic support                                                                                                                    |

| Proposed Solution |                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach          | Utilize machine learning algorithms to analyze behavioral engagement patterns (e.g., login frequency, video watch time, assignment submission timeliness) and predict academic performance outcomes. |
| Key Features      | Real-time engagement tracking, automated early warning alert systems for at-risk students, predictive performance scoring, and actionable dashboard visualizations.                                  |

### Resource Requirements

| Resource Type | Description             | Specification/Allocation                                                       |
|---------------|-------------------------|--------------------------------------------------------------------------------|
| Hardware      |                         |                                                                                |
| Hardware      | Computing Resources     | Multi-core CPU / GPU (e.g., 2 x NVIDIA V100 GPUs)                              |
| Hardware      | Memory                  | 8 GB RAM (minimum)                                                             |
| Hardware      | Storage                 | 1 TB SSD for raw logs, models, and analytical outputs                          |
| Software      |                         |                                                                                |
| Software      | Frameworks              | Python web/application framework (e.g., Flask, Streamlit)                      |
| Software      | Libraries               | Data processing & ML libraries (e.g., scikit-learn, pandas, numpy, matplotlib) |
| Software      | Development Environment | IDE & Version Control (e.g., Jupyter Notebook, VS Code, Git)                   |

| Data |             |                                                                                                                                                 |
|------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Data | Data Source | Educational LMS interaction dataset (e.g., Kaggle LMS dataset / Open University Analytics Dataset, ~10,000+ student records in CSV/JSON format) |